

Post-quantum transition for device bounded anonymous credentials: a EUDI wallet application

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Abstract

In 2026, member states of the European Union are expected to have running their own EUDI wallet applications instances compliant with the ARF architecture guidelines. The goal of the ARF architecture is to provide privacy preserving anonymous credentials to users, but the current implementation fails.

In this thesis we will explore the current state of the art and give an overview of current implementations of anonymous credentials. We will then explore the post-quantum settings which has not been faced by the current implementations and we will give an important result which allowed us to implement efficient privacy preserving post-quantum anonymous credentials, preserving legacy architectures in their almost entirety.

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1 Introduction

The european union on 11 April 2024 has published a roadmap for the post quantum digital transition of members digital infrastructures.

1.1 Overview

2 State of the art

2.1 Anonymous Credentials

2.2 Post quantum anonymous credentials

3 Implementation